4. Swift Collections

Task 1:

Code:

```
import Foundation
var subjects = ["Swift", "UIKit", "SwiftUI"]
subjects.append("Networking")
subjects.insert("Testing", at: 1)
subjects.remove(at: subjects.firstIndex(of: "UIKit")!)
print("Sorted subjects:", subjects.sorted())
var registered: Set<String> = ["Deepika", "Brinda", "Ranisha"]
registered.insert("Deepika")
registered.insert("Hema")
print("Registered students:", registered)
print("Number of students:", registered.count)
var marks: [String: Int] = [
 "Deepika": 85,
 "Brinda": 92,
 "Ranisha": 78 ]
marks["Hema"] = 88
marks["Deepika"] = 95
print("Esha's mark:", marks["Esha"] as Any)
for name in marks.keys.sorted() {
 print("\(name): \(marks[name]!)")
}
if let highest = marks.max(by: { $0.value < $1.value }) {
 print("Highest mark:", highest.value)
 print("Top student:", highest.key)
}
```

OUTPUT:

```
Output

Sorted subjects: ["Networking", "Swift", "SwiftUI", "Testing"]
Registered students: ["Deepika", "Ranisha", "Brinda", "Hema"]
Number of students: 4
Esha's mark: nil
Brinda: 92
Deepika: 95
Hema: 88
Ranisha: 78
Highest mark: 95
Top student: Deepika
```

Task 2:

Code:

```
import Foundation

let sentence = "Tomato is a fruit and Tomato is also a vegetable"

let words = sentence.split(separator: " ").map { String($0) }

print("Total words:", words.count)

let uniqueWords = Set(words)

print("Unique words:", uniqueWords.sorted())

var frequency: [String: Int] = [:]

for word in words {

 frequency[word, default: 0] += 1

}

print("Word frequencies:")

for word in frequency.keys.sorted() {

 print("\(word): \(frequency[word]!)")

}

let rankedWords = frequency.sorted {

 if $0.value != $1.value {
```

```
 return $0.value > $1.value
 }
 return $0.key < $1.key
}
print("Ranked words:")
for (word, count) in rankedWords {
 print("\(word): \(count)")
}
```

OUTPUT:



Task 3:

Code:

```
import Foundation
let swiftClub: Set<String> = ["Brinda", "Deepika", "Hema", "Ranisha"]
let uiClub: Set<String> = ["Hema", "Ranisha", "Esha"]
let both = swiftClub.intersection(uiClub)
print("Both clubs:", both.sorted())
let onlySwift = swiftClub.subtracting(uiClub)
print("Only Swift club:", onlySwift.sorted())
```

```
let everyone = swiftClub.union(uiClub)

print("Everyone:", everyone.sorted())

let exactlyOne = swiftClub.symmetricDifference(uiClub)

print("Exactly one club:", exactlyOne.sorted())
```

OUTPUT:

Output

```
Both clubs: ["Hema", "Ranisha"]
Only Swift club: ["Brinda", "Deepika"]
Everyone: ["Brinda", "Deepika", "Esha", "Hema", "Ranisha"]
Exactly one club: ["Brinda", "Deepika", "Esha"]
```